

Abdul Rahim

AI Engineer | Retrieval, Agents & Local Inference

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SUMMARY

Software engineer building AI applications with Python, TypeScript, and AWS. Experience spans structured company research, hybrid retrieval, browser agents, and streaming speech inference on Apple Silicon.

EXPERIENCE

Software Engineer | Consultadd

March 2025 - March 2026

Pune

- Built BizInsights, a seven-stage research workflow for market analysis, competitors, SWOT, and value propositions.
- Built web crawling and iterative URL discovery, then validated LLM-generated research outputs using Pydantic schemas.
- Built FastAPI endpoints and status tracking with DynamoDB and S3; implemented Google OAuth and role-based access.
- Worked on AWS voice campaign orchestration, concurrency control, and post-call reporting using Lambda, SQS, DynamoDB, and EventBridge.

PROJECTS

Nemotron Streaming ASR

[GitHub](#)

Python, MLX, FastConformer-RNNT

- Implemented cache-aware streaming for NVIDIA's Nemotron ASR model on Apple Silicon, preserving encoder/decoder state across audio chunks.
- Added bounded audio history, reference-equivalence tests, a latency benchmark, and a macOS dictation app.

PDF RAG API

[GitHub](#)

Python, FastAPI, LlamaIndex, Qdrant, Docker

- Combined dense embeddings and BM25 with reciprocal rank fusion and BGE reranking for a FastAPI PDF question-answering service.
- Added asynchronous ingestion, content-hash deduplication, and SSE answer streaming with source references and stage latency metrics.

ApplyPilot AI

[GitHub](#)

TypeScript, Python, Playwright, FastAPI, LangGraph, MCP

- Connected a LangGraph job-discovery agent to Playwright handlers for Greenhouse, Lever, SmartRecruiters, and Workday-style forms.
- Added an OpenAI Agents SDK fallback through Playwright MCP and SSE task monitoring in FastAPI.

TECHNICAL SKILLS

Languages & backend: Python, TypeScript, JavaScript, FastAPI, REST APIs, Pydantic

AI applications: OpenAI Agents SDK, LangGraph, MCP, structured outputs, tool calling

Retrieval & inference: Retrieval-Augmented Generation (RAG), vector embeddings, hybrid search, cross-encoder reranking, LlamaIndex, Qdrant, MLX

Cloud & delivery: AWS (EC2, Lambda, SQS, DynamoDB, S3, EventBridge), vLLM, Docker, GitHub Actions, pytest

EDUCATION

B.Tech in Computer Science | SRM University, Guntur

Minor in AI/ML | CGPA: 8.86/10

2021 - 2025

DSA & ACHIEVEMENTS

- Solved 700+ algorithmic problems across [LeetCode](#), [CodeChef](#), and GeeksForGeeks.
- Earned a 3-star rating on [CodeChef](#).
- Secured 2nd place in SRM CodeClash.